

Mehmood Ahmad

587-568-3734 | mehmood3@ualberta.ca | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Alberta

Bachelor of Science - Computer Science, Minor in Biological Sciences

Edmonton, AB

Jan 2027

EXPERIENCE

University of Alberta

Software Engineer Intern

Edmonton, AB

May 2024 – September 2025

- Led development of a Safety Monitor System combining a **YOLOv11** detector with vision-language model reasoning for real-time hazard detection on live camera feeds, improving hazard-identification F1-score from 34.5% to 50.6% with only ~2.5ms added inference latency.
- Built a high-performance **FFmpeg** streaming pipeline and a VR-controlled Remote Inspection Robot using **ROS** for real-time navigation, paired with a companion Unity/C# VR interface.
- Automated deployment and system startup with Bash scripting, reducing multi-step manual workflows to a single command.

Aro Robotic Systems

Software Engineer Intern

Edmonton, AB

May 2023 – September 2023

- Cut average deployment time by ~**70%** by designing containerized pipelines with Docker and Jenkins across staging and production.
- Implemented end-to-end CI/CD workflows in **GitLab CI** with automated testing, linting, and security checks to catch issues earlier.
- Lowered operational costs by ~**15%** by provisioning and right-sizing cloud infrastructure on **AWS** and DigitalOcean.

Aro Robotic Systems

Software Engineer Intern

Edmonton, AB

May 2022 – September 2022

- Designed and built the company website arosystems.ai from scratch using WordPress, delivering a fully responsive web presence for the startup.
- Customized the WordPress theme with reusable HTML/CSS/JavaScript components (navigation, landing sections, cards, contact forms) matching brand guidelines.

PUBLICATIONS

- M. Adil, **M. Ahmad**, et al., “Integration of Object Detection and Small VLMs for Construction Safety Hazard Identification,” *arXiv:2604.05210*, 2026.

PROJECTS

Trackme | React Native, TypeScript, FastAPI, SQLite, Apple HealthKit

[GitHub Link](#)

- Built a full-stack health tracker with a **React Native** (TypeScript) frontend and **FastAPI**/SQLite backend, integrating Apple HealthKit for real-time activity data.
- Designed an AI insights engine that analyzes habit patterns and automatically looks up meal nutrition via the Nutritionix/Edamam APIs to deliver personalized recommendations.

Offspring.exe | React, TypeScript, Vite

[GitHub Link](#)

- Built a genetics simulation modeling Mendelian inheritance and polygenic traits through a seeded Monte Carlo simulation of 20,000 possible offspring.
- Designed a procedural SVG avatar generator rendering the most likely outcome directly from simulation data, applying genetics coursework from my Biological Sciences minor.

Event Lottery System Application | Android, Firebase, QR Code, Java, Jetpack Compose

[GitHub Link](#)

- Developed an Android event sign-up app in Java/Jetpack Compose using a fair lottery system with entrant, organizer, and admin roles.
- Implemented **Firebase**-backed real-time storage with QR code registration, scanning, and geolocation verification.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, Java, SQL (SQLite3, MongoDB), JavaScript, HTML/CSS, Shell

Frameworks & Libraries: Node.js, Django, Bootstrap, Unity, ROS, pandas, NumPy, Matplotlib, transformers, PEFT, Tkinter, BeautifulSoup, Scrapy, Torch

Cloud & DevOps: Docker, Jenkins, GitLab CI/CD, AWS, DigitalOcean, Git, Bash

Developer Tools: VS Code, Visual Studio, PyCharm, WebStorm, REST APIs

Technologies: Large Language Models (LLMs), VLMs, Computer Vision, YOLOv11, FFmpeg